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0.0.1 Siegel Modular Forms and Igusa cusp form Φ_{10}

A Siegel modular form of weight k is a holomorphic function $F : \mathbb{H}_g \rightarrow \mathbb{C}$ satisfying

$$f(M \cdot \tau) = \det(c\tau + d)^k f(\tau) \quad (1)$$

$\forall M = (a, b; c, d) \in Sp(2g, \mathbb{Z})$, $\tau \in \mathbb{H}_g$ and if $g = 1$ then f is holomorphic at the cusps. Here $\mathbb{H}_g$ is the Siegel upper half space of genus g defined as

$$\mathbb{H}_g = \{\tau \in M_{g \times g}(\mathbb{C}) | \tau^T = \tau, \Im(\tau) > 0\} \quad (2)$$

One such Siegel modular form is the Igusa cusp form Φ_{10} which obeys the following transformation properties

$$\begin{aligned} \Phi_{10}\left((A\Omega + B)(C\Omega + D)^{-1}\right) &= \det(C\Omega + D)^{10} \Phi_{10}(\Omega) \\ \text{where } \Omega &= \begin{bmatrix} \rho & v \\ v & \sigma \end{bmatrix} \quad \begin{bmatrix} A & B \\ C & D \end{bmatrix} \in Sp(2, \mathbb{Z}) \\ \forall \Lambda \in Sp(2, \mathbb{Z}) &\text{ satisfies} \\ \begin{bmatrix} A & B \\ C & D \end{bmatrix}^T \begin{bmatrix} 0 & I_2 \\ -I_2 & 0 \end{bmatrix} \begin{bmatrix} A & B \\ C & D \end{bmatrix} &= \begin{bmatrix} 0 & I_2 \\ -I_2 & 0 \end{bmatrix}, \quad I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}. \end{aligned}$$

Φ_{10} is invariant under S-duality transformations ($SL(2, \mathbb{Z})$) transformations operating on Ω as

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}_S = \begin{bmatrix} a & -b & 0 & 0 \\ -c & d & 0 & 0 \\ 0 & 0 & d & c \\ 0 & 0 & b & a \end{bmatrix} \quad (3)$$

Thus we have

$$\rho' = \rho - \frac{\gamma v^2}{\gamma\sigma + \delta} \quad \sigma' = \frac{\alpha\sigma + \beta}{\gamma\sigma + \delta} \quad v' = \frac{v}{\gamma\sigma + \delta}.$$

$\forall \lambda, \mu, v \in \mathbb{Z}$

$$\Phi_{10}(\rho + \lambda, \sigma + \mu, v + \nu) = \Phi_{10}(\rho, \sigma, v) \quad (4)$$

Thus Φ_{10} has a period of 1, therefore there exists a fourier expansion. So we have a net transformation

$$\begin{aligned} \Phi_{10}(\rho', \sigma', v') &= (\gamma\sigma + \delta)^{10} \Phi_{10}(\rho, \sigma, v) \\ \rho' &= a^2(\rho - \chi) + b^2\kappa - 2ab\psi \\ \sigma' &= c^2(\rho - \chi) + d^2\kappa - 2cd\psi \\ v' &= -ac(\rho - \chi) - bd\kappa + (ad + bc)\psi - \Sigma \\ \chi &= \frac{\gamma v^2}{\gamma\sigma + \delta} \quad \kappa = \frac{\alpha\sigma + \beta}{\gamma\sigma + \delta} \quad \Psi = \frac{v}{\gamma\sigma + \delta} \end{aligned}$$

Since $v = 0$ we get the poles given by

$$n_2(\rho\sigma - v^2) + jvn_1\sigma - m_1\rho + m_2 = 0 \quad (5)$$

$\forall z := \{m_1, m_2, j, n_1, n_2\} \in \mathbb{Z}$, where we have the constraint that

$$m_1 n_1 m_2 n_2 = \frac{1}{4}(1 - j^2) \quad (6)$$

and the poles with the constraint is invariant under $z \rightarrow -z$ thus we strictly consider $n_2 \geq 0$. [?].
As proven in [?], poles 5 with $n_2 \geq 1$ can be parametrised by nine integers

$$\begin{aligned} n_2 &= -ac\gamma, \\ j &= ad + bc \\ n_1 &= -bd\alpha - \gamma\Sigma \\ m_1 &= ac\delta \\ m_2 &= -bd\beta - \delta\Sigma \end{aligned}$$

where eight of these can be arranged into $\Lambda_i \in SL(2, \mathbb{Z})$ $i = 1, 2$

$$\Lambda_1 = \begin{bmatrix} \alpha & \beta \\ \gamma & \delta \end{bmatrix} \quad \Lambda_2 = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

with $a > 0$, $c < 0$, $\gamma > 0$, $\alpha \in \mathbb{Z}/\gamma\mathbb{Z}$ and $\Sigma \in \mathbb{Z}$.

One can use the new coordinates for all of the poles restricted to $n_2 \geq 1$ we have a corresponding element of the set $P \cup \{\Sigma \in \mathbb{Z}\}$,

$$P := \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \begin{bmatrix} \alpha & \beta \\ \gamma & \delta \end{bmatrix} \in SL(2, \mathbb{Z}) \mid \alpha, \gamma > 0, c < 0, \alpha \in \mathbb{Z}/\gamma\mathbb{Z} \right\} \quad (7)$$

We have the sets of matrices

$$S_\Gamma = \left\{ \begin{bmatrix} \alpha & \beta \\ \gamma & \delta \end{bmatrix} \in SL(2, \mathbb{Z}), \gamma > 0 \right\} \quad S_G = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in SL(2, \mathbb{Z}) \mid a > 0, c < 0 \right\} \quad (8)$$

$$\Gamma_\infty = \left\{ \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix} \in SL(2, \mathbb{Z}), k \in \mathbb{Z} \right\} \quad (9)$$

We sum over the elements of Γ_∞/S_Γ , S_G and $\{\Sigma \in \mathbb{Z}\}$.

Now if we look at the residues at $v' = 0$

we can let

$$\Lambda(\sigma, v) = \frac{\gamma v^2}{\gamma\sigma + \delta} - \frac{bd}{ac} \left(\frac{a\sigma + \beta}{\gamma\sigma + \delta} \right) + \frac{(ad + bc)}{ac} \frac{v}{\gamma\sigma + \delta} - \frac{1}{ac} \Sigma$$

by ?? we have $v' = -ac(\rho - \Lambda(\sigma, v))$

Should $\Lambda = \rho$ then if $\Sigma \rightarrow \Sigma + ac$, $\implies \Lambda \rightarrow \Lambda - 1$ then ρ_1 is shifted by an integer which then is no longer in its fundamental domain $[0, 1)$ integral domain. So we add the restriction that $\Sigma \in \mathbb{Z}/|ac|\mathbb{Z}$.

$$\Phi_{10}^{-1} \Big|_{v \rightarrow 0} \rightarrow -\frac{1}{4\pi^2} \frac{1}{v^2} \frac{1}{\eta^{24}(\rho)} \frac{1}{\eta^{24}(\sigma)} \quad (10)$$

We have

$$\frac{d}{d\tau} \log \eta^{24}(\tau) = 2\pi i E_2(\tau) \quad \text{Res}_{\rho=\Lambda} f(\rho) = \lim_{\rho \rightarrow \Lambda} \frac{\partial}{\partial \rho} \left((\rho - \Lambda)^2 f(\rho) \right) \quad (11)$$

$$(-1)^{\ell+1} \frac{(\gamma\sigma + \delta)^{10}}{ac} \left(\frac{m}{ac} + \frac{a}{c} E_2(\rho'_*) + \frac{c}{a} E_2(\sigma'_*) \right) \frac{1}{\eta^{24}(\rho'_*)} \frac{1}{\eta^{24}(\sigma'_*)} e^{-2\pi i(m\Lambda + n\sigma + \ell v)} \quad (12)$$

with

$$\begin{aligned} \rho'_* &= -\frac{b}{c} \left(\frac{a\sigma + \beta}{\gamma\sigma + \delta} \right) + \frac{a}{c} \left(\frac{v}{\gamma\sigma + \delta} \right) - \frac{a}{c} \Sigma \\ \sigma'_* &= \frac{d}{a} \left(\frac{a\sigma + \beta}{\gamma\sigma + \delta} \right) - \frac{c}{a} \left(\frac{v}{\gamma\sigma + \delta} \right) - \frac{c}{a} \Sigma \\ v'_* &= 0. \end{aligned}$$

Thus we get

$$-2\pi i \lim_{\rho \rightarrow \Lambda(\sigma, v)} (-1)^{\ell+1} \frac{\partial}{\partial \rho} \left((\rho - \Lambda(\sigma, v))^2 (\gamma\sigma + \delta)^{10} \frac{-1}{4\pi^2} \frac{1}{v'^2} \frac{1}{\eta^{24}(\rho')} \frac{1}{\eta^{24}(\sigma')} e^{-2\pi i(m\rho + n\sigma + \ell v)} \right) \quad (13)$$

$$d(m, n, \ell)_{\Delta > 0} = \sum_P \sum_{\Sigma \in \mathbb{Z}/|ac|\mathbb{Z}} \frac{(-1)^{\ell+1}}{\gamma^2(ac)^{13}} \int_{\hat{C}} \frac{d\tau \wedge d\bar{\tau}}{(\tau - \bar{\tau})^{13}} \left(\frac{m}{ac} + \frac{a}{c} E_2(\rho'_0) + \frac{c}{a} E_2(\sigma'_0) \right) \cdot \frac{1}{\eta^{24}(\rho'_0)} \frac{1}{\eta^{24}(\sigma'_0)} e^{-2\pi i \Lambda} \quad (14)$$

0.0.2 Eisenstein Series

The Eisenstein series for $SL(2, \mathbb{Z})$ is given by

$$E_k = \frac{1}{2} \sum_{c, d \in \mathbb{Z}} \frac{1}{(c\tau + d)^k} \quad (c, d) = 1 \quad (15)$$